



There are restrictions we want to keep

3

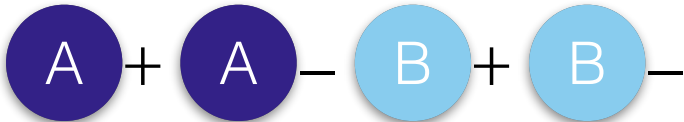
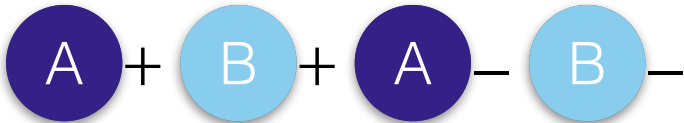
9

```
def update(account: cown[Account], amount: int) {  
  when(account) { A  
    if(account.balance + amount > 0) {  
      account.balance += amount  
      val id = account.id()  
      when(log) { B log.record(id, amount) }  
    } } }  
}
```



Perminituted

upd(+ -), log(+ -)

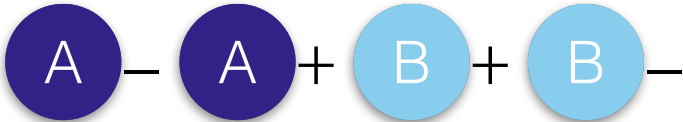


order

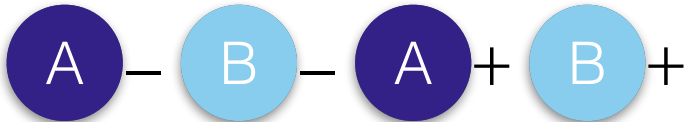
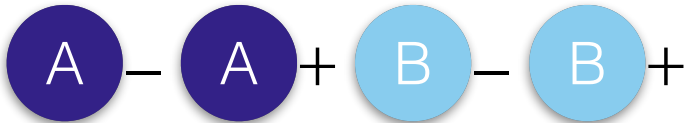
$\text{upd}(+-), \log(-+)$



upd(-+), log(+ -)



upd(-+), log(-+)













In this case, the semantics provide the appropriate restrictions.

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```
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      val id = account.id()  
      when(log) { B log.record(id, amount) }  
    }  
  }  
}
```

In this case, the semantics provide the appropriate restrictions.

	upd(+ -), log(+ -)	upd(+ -), log(- +)	upd(- +), log(+ -)	upd(- +), log(- +)
Order	A + B + A - B - A + A - B + B -	A + A - B - B +	A - A + B + B -	A - A + B - B + A - B - A + B +
Permitted	✓	✗	✗	✓



Insight
